MedQ-Deg: A Multidimensional Benchmark for Evaluating MLLMs Across Medical Image Quality Degradations

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Abstract

Despite the impressive performance on standard benchmarks, multimodal large language models face critical challenges in real-world clinical environments where medical images inevitably suffer various degradations, *e.g.*, noise, artifacts, resolution loss, *etc.* Existing benchmarks exhibit two key limitations: (1) absence of large-scale multidimensional assessment across medical image quality gradients and (2) no systematic confidence calibration analysis. To address these gaps, we present **MedQ-Deg**, a systematic benchmark for comprehensively evaluating medical MLLM across image quality gradients. MedQ-Deg offers multi-dimensional evaluation across 18 distinct medical image degradations, 30 fine-grained capabilities, and 7 imaging modalities. Each degradation is implemented at 3 severity levels calibrated by human experts. We curate 24,894 question-answer pairs from three top-tier benchmarks. Our evaluation of 40 mainstream MLLMs across capability dimensions, degradation categories, and imaging modalities reveals systematic performance degradation as image quality degrades. More critically, we demonstrate the **AI Dunning-Kruger Effect**: models maintain inappropriately high confidence despite severe accuracy collapse. These findings underscore the urgent need for enhanced reliability under image quality variations before clinical deployment.

1. Introduction

Multimodal Large Language Models (MLLMs) have achieved remarkable progress in medical vision-language tasks. General-purpose (Hurst et al., 2024; Comanici et al., 2025; Anthropic, 2025) and medical-specialized ones (Jiang et al., 2025; Sellergren et al., 2025) have approached or surpassed human expert performance on standard medical benchmarks (Ye et al., 2024; Hu et al., 2024), showing immense potential for assisting in diagnostic workflows and automating clinical reporting. These benchmarks, however, predominantly evaluate models on carefully curated, high-quality medical images under controlled conditions. Such evaluation paradigms often mask a critical vulnerability: the lack of reliable performance when faced with the imperfect reality of clinical imaging.

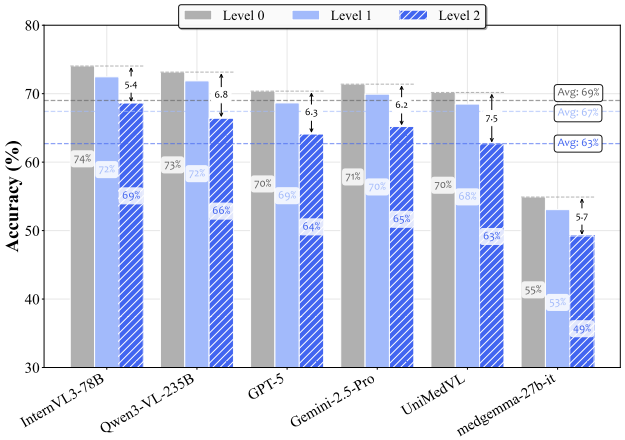

Figure 1. **Severity-level degradation trends for different models on MedQ-Deg benchmarks.** Please see Figure 4 for full results.

This vulnerability becomes evident when we consider real-world clinical environments that are far more complex than standard benchmark datasets suggest. Medical images in nature suffer from various degradations, including noise from low-dose scanning, motion artifacts from patient movement, field inhomogeneity from aging equipment, and illumination variations in endoscopy, particularly in low-resource settings or smaller clinics where advanced imaging equipment is unavailable. As shown in Figure 1, models exhibit consistent accuracy decline across increasing degradation severity levels. While human physicians can usually navigate these imperfections and still reach a correct diagnosis, the reliability of MLLMs in such scenarios remains largely unproven, creating a significant barrier to safe deployment in healthcare.

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More alarmingly, our preliminary investigations reveal a systematic pattern of metacognitive failure in MLLMs when confronted with degraded medical images. Models not only suffer accuracy drops but also exhibit a striking inability to recognize the boundaries of their own competence, maintaining inappropriately high confidence even when producing erroneous predictions on corrupted inputs. As illustrated in Figure 2, an MLLM correctly detected the liver lesion in a computed tomography (CT) image when no degradation is present; however, when the same image is degraded (*e.g.*, with sparse-view CT artifacts due to the subsampling of raw measurement data (Jin et al., 2017)), the model yields false-negative predictions while assigning similarly high confidence. This disconnect between capability and self-assessment manifests across diverse degradations and severity levels, suggesting a fundamental deficit in model awareness rather than isolated failures.

We term this phenomenon the **AI Dunning-Kruger Effect**, drawing an analogy to the well-documented cognitive bias in human psychology where individuals with limited ability overestimate their competence (Kruger & Dunning, 1999). This metacognitive blindness, that is, the inability to "know what it doesn't know," poses particularly severe risks in clinical deployment, as overconfident erroneous inferences may prevent the activation of necessary human oversight and mislead physicians into trusting unreliable AI recommendations.

Despite these critical vulnerabilities, existing evaluation frameworks remain inadequate for systematically assessing model performance under image quality variations. Current benchmarks suffer from two fundamental limitations. (1) *Absence of large-scale multidimensional assessment under medical image degradations*. General image quality benchmarks (Li et al., 2025b; Qiu et al., 2025) typically focus on natural image corruptions (*e.g.*, Gaussian noise, blur, JPEG compression) that fail to capture medical imaging-specific degradations such as motion artifacts in magnetic resonance imaging (MRI). Meanwhile, existing medical MLLM benchmarks (Hu et al., 2024; Ye et al., 2024) are primarily designed for clean images without evaluation across different image degradation types and severity degrees. Although recent work (Cheng et al., 2025; XU et al., 2025) has begun exploring medical MLLM performance under degradations, they lack the scale and comprehensiveness necessary for systematic evaluation across diverse modalities and degradations. (2) *No systematic confidence calibration analysis*. Existing medical benchmarks provide only coarse-grained overall accuracy metrics, without quantifying model confidence calibration and metacognitive awareness under degradation, which are precisely the dimensions essential for detecting AI Dunning-Kruger behaviors and ensuring safe clinical deployment.

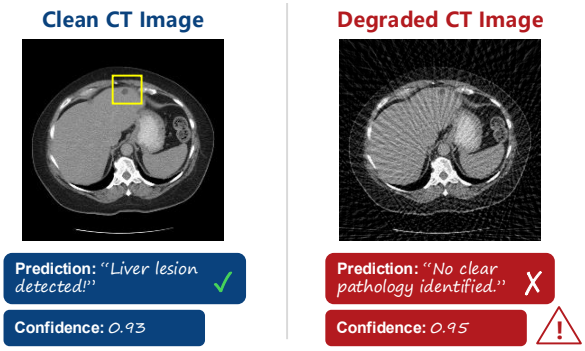

Figure 2. **Illustration of AI Dunning-Kruger Effect.** An MLLM correctly identifies a liver lesion in the clean CT image but yields erroneous predictions with similarly high confidence when the image is corrupted while the lesion is still visible. The bounding box is only for visualization and is not provided to the MLLM.

To address these gaps, we propose **MedQ-Deg**, a benchmark for comprehensively evaluating medical MLLM robustness against clinically realistic image degradations. As shown in Figure 3. Our benchmark is designed to enable both fine-grained capability assessment under different medical imaging degradations and confidence calibration analysis, addressing precisely the dimensions missing from existing evaluation frameworks. Our contributions are threefold:

(1) A systematic benchmark with hierarchical evaluation framework. We construct MedQ-Deg, comprising 24,894 question-answer pairs curated from multiple established medical datasets (Hu et al., 2024; Ye et al., 2024; Zuo et al., 2025). The benchmark features a three-tier framework along two orthogonal axes: *capability* (spanning from high-level categories through 6 clinical tasks to 30 fine-grained skills) and *degradation* (covering 5 major categories and 18 subcategories across 7 imaging modalities, with each instantiated at 3 severity levels validated by expert radiologists). This hierarchical design enables structured and clinically meaningful assessment across image quality degradation.

(2) Quantitative evidence of the AI Dunning-Kruger Effect in medical MLLMs. We systematically analyze the confidence-performance relationship under degradation and provide large-scale empirical evidence of AI Dunning-Kruger Effect: medical MLLMs remain markedly overconfident even as their true capabilities deteriorate. This metacognitive failure reveals that current models lack the self-awareness required for safe clinical deployment.

(3) Comprehensive evaluation across models and dimensions. We conduct extensive evaluation of 40 mainstream MLLMs spanning proprietary general models, open-source general models, and medical-specialized models. Our multidimensional analysis examines performance degradation across capability dimensions, degradation categories, and imaging modalities, providing the most comprehensive char-

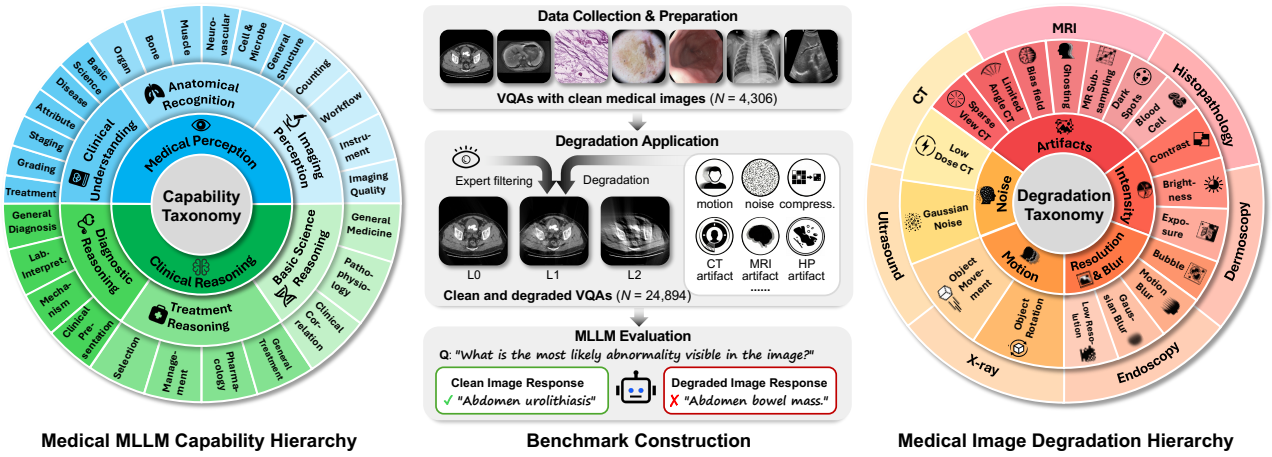

Figure 3. **Benchmark framework overview. Left:** Capability taxonomy spanning Medical Perception and Clinical Reasoning with mid-level tasks and fine-grained skills. **Middle:** Data pipeline from clean medical images to degraded samples via controlled corruption at severity degrees L0–L2, with human filtering to ensure clinical realism and relevance. **Right:** Degradation taxonomy organized by category (artifacts, noise, intensity jitter, motion interference, resolution & blur issues) and modality-specific variants.

acterization of medical MLLM behavior under image quality variations to date.

2. The MedQ-Deg Benchmark

MedQ-Deg benchmark is designed to evaluate how image quality degradation affects the diagnostic performance and metacognitive reliability of existing MLLMs. As illustrated in Figure 3, the benchmark is built upon a dual-tier hierarchy that stratifies both model capabilities (Section 2.1) and clinical degradation patterns (Section 2.2), supported by a human-curated data pipeline that ensures clinical realism (Section 2.3) and a novel suite of metacognitive metrics designed to quantify model overconfidence under stress (Section 2.4). By doing so, we move beyond simple accuracy scores to provide multi-dimensional performance assessment and fine-grained capability characterization.

2.1. Medical MLLM Capability Hierarchy

A comprehensive evaluation requires more than just a single binary pass-fail metric. We organize the capability tier into a three-stage hierarchy that reflects the cognitive progression of a human clinician (Figure 3, left). This hierarchical design enables fine-grained analysis of model strengths and weaknesses across different levels of medical cognition:

1) High-level Capabilities ($\mathcal{T}^{(1)}$): We partition the capability space into two fundamental categories: *medical perception* and *clinical reasoning*. This distinction is critical because robustness often breaks down unevenly; a model might retain the ability to see a lesion or recognize the surgical workflow (perception) while losing the ability to derive the complex pathologies and the clinical significance (reasoning), or vice versa.

2) Mid-level Tasks ($\mathcal{T}^{(2)}$): We decompose each high-level capability into six clinically relevant dimensions. Perception is subdivided into *anatomical recognition, imaging perception, and clinical understanding*, focusing on identifying structures and descriptive features. Reasoning is subdivided into *basic science reasoning, treatment reasoning, and diagnostic Reasoning*, which test the model's ability to synthesize visual evidence with medical logic.

3) Fine-grained Skills ($\mathcal{T}^{(3)}$): Each mid-level task is further stratified into specific clinical skills, yielding 30 fine-grained capabilities that span the full spectrum of medical image understanding competencies. The complete capability taxonomy with detailed task descriptions is provided in Appendix A.

2.2. Medical Image Degradation Hierarchy

Clinical images are rarely perfect. We model this reality by applying 18 distinct degradation variants across five fundamental categories: intensity jitter, artifacts, noise, motion interference, and resolution & blur issues, involving seven imaging modalities including CT, MRI, histopathology, dermoscopy, endoscopy, X-ray, and ultrasound, as shown in the right panel of Figure 3.

Unlike general benchmarks that mainly use common image corruptions (*e.g.*, Gaussian noise), MedQ-Deg involves modality-specific degradations that reflect real-world hardware and patient-related constraints. We define a degradation operation $\mathcal{C}_i^s$ mathematically as:

$$\mathcal{I}_c = \mathcal{C}_i^s(\mathcal{I}_0; \theta_i^s), \tag{1}$$

where $\mathcal{I}_0$ denotes an original diagnostic-quality image, $i \in \{1, 2, ..., N\}$ indexes the degradation variants, and

$s \in \{0, 1, 2\}$ represents the severity degree. We use three severity degrees calibrated by expert radiologists. L0 represents the original clean image, L1 introduces mild degradation, and L2 applies severe degradation that challenges human interpretation but still preserves enough critical information for a possible diagnosis.

2.3. Dataset Construction

We curate MedQ-Deg by aggregating data from three well-known medical MLLM benchmarks: GMAI-MMBench (Ye et al., 2024), OmniMedVQA (Hu et al., 2024), and MedXpertQA (Zuo et al., 2025), which together provide comprehensive coverage of medical modalities, task types, and capability dimensions. For each deduplicated visual question-answer (VQA) pair, we randomly apply one to three modality-specific degradations at severity degrees L1 and L2. To maintain clinical validity, we implemented a human-in-the-loop filtering process, where medical experts review both images and questions to ensure that the applied corruptions do not completely erase the diagnostic features required to answer the associated questions, and that the question itself is clinically relevant and not trivially answerable from superficial cues. Detailed dataset statistics are provided in Table 1, Table 4 (in Appendix A) and Table 5 (in Appendix B)

Table 1. **Dataset statistics.** Summary of MedQ-Deg composition across modalities, capabilities, degradation variants, and severity-degree splits.

Attribute	Value
Total QA pairs	24,894
Medical imaging modalities	7
Fine-grained capabilities	30 (across 6 mid-level tasks)
Degradation variants	18 (9 general + 9 medical-specific)
L0 (Clean image without degradatioon)	4,306
L1 (Mild degradation)	10,294
L2 (Severe degradation)	10,294

2.4. Evaluation Metrics

To assess model robustness and metacognitive capabilities under image degradation, we utilize a combination of accuracy and uncertainty metrics for quantifying both actual performance and calibration shift.

Actual Performance. For each VQA pair in our dataset, models generate predictions through multiple-choice selection. We evaluate performance using accuracy as the primary metric:

$$\text{Acc}(f, \mathcal{D}) = \frac{1}{|\mathcal{D}|} \sum_{x \in \mathcal{D}} \mathbb{I}\big[\hat{y}(x) = y^*(x)\big], \tag{2}$$

where $\mathcal{D}$ denotes the evaluation dataset, $y^*(x)$ represents the ground-truth answer, $\hat{y}(x)$ is the model's prediction, and $\mathbb{I}[\cdot]$ is the indicator function.

Perceived Confidence. We measure model confidence through voting-based prediction consistency. For each sample, we conduct T independent inference trials. The voting distribution is $p_i(x) = \frac{1}{T}\sum_{t=1}^{T} \mathbb{I}\big[y^{(t)} = i\big]$ for $i \in \{1, \ldots, K\}$, where K is the number of answer choices and $y^{(t)}$ denotes the prediction in trial t. We quantify prediction uncertainty using normalized entropy: $H(x) = -\sum_{i=1}^{K} p_i(x) \log p_i(x)$, with maximum entropy $H_{\max} = \log K$ for complete uncertainty. Sample-level confidence is defined as the inverse of normalized entropy:

$$C(x) = 1 - H(x)/\log K, \tag{3}$$

where $C(x) \in [0, 1]$, with $C(x) = 1$ indicating perfect prediction consistency and $C(x) \approx 0$ for high uncertainty.

Calibration Shift. Model calibration shift on dataset $\mathcal{D}$ is quantified as the gap between perceived confidence and actual accuracy:

$$\Delta_{\text{calib}}(f, \mathcal{D}) = \frac{1}{|\mathcal{D}|} \sum_{x \in \mathcal{D}} C(x) - \text{Acc}(f, \mathcal{D}). \tag{4}$$

A positive Δ_{calib} indicates overconfidence (model overestimates its capability), zero represents well-calibrated confidence, and negative values indicate underconfidence.

We use Δ_{calib} to characterize two distinct forms of the AI Dunning-Kruger Effect (DKE). *Intra-Model DKE* occurs when a model's accuracy collapses as image uality drops from L0 to L2, but the calibration shift remains stable or increases, defined as:

$$\text{Acc}(f, \mathcal{D}_{\text{L0}}) > \text{Acc}(f, \mathcal{D}_{\text{L2}}) \ \wedge \ \Delta_{\text{calib}}(f, \mathcal{D}_{\text{L0}}) \leq \Delta_{\text{calib}}(f, \mathcal{D}_{\text{L2}}), \tag{5}$$

where $\mathcal{D}_{\text{L0}}$ and $\mathcal{D}_{\text{L2}}$ denote evaluation sets at L0 and L2 severity degree respectively. This reveals a metacognitive failure where the model is "unaware" of its own declining capability. *Inter-Model DKE* describes a trend that lower-performing models show higher calibration shift compared to higher-performing models:

$$\text{Acc}(f_1, \mathcal{D}) < \text{Acc}(f_2, \mathcal{D}) \ \wedge \ \Delta_{\text{calib}}(f_1, \mathcal{D}) > \Delta_{\text{calib}}(f_2, \mathcal{D}). \tag{6}$$

3. Experiments

This section presents a comprehensive evaluation of 40 representative MLLMs. We start by detailing our experimental setup (Section 3.1), and then organize our analysis into the following perspectives: performance across severity degrees (Section 3.2), capability dimension analysis (Section 3.3), degradation type sensitivity (Section 3.4), modality-specific performance (Section 3.5), and overconfidence under medical image degradations (Section 3.6).

3.1. Experimental Setup

We evaluate 40 representative MLLMs spanning three groups: (1) **9 *Proprietary general models***, including GPT-

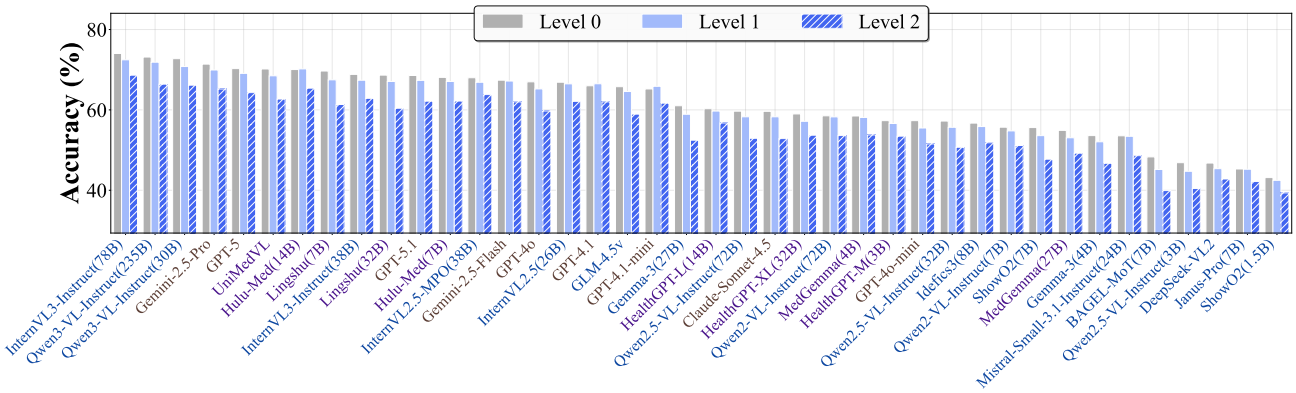

Figure 4. **Model performance across medical image quality degradation.**

Table 2. **Capability-level performance (L1/L2) with degradation magnitude.** Average accuracy across the six mid-level capability dimensions, computed over L1 and L2 degradations. Numbers in parentheses show performance drop from L0 to L1&L2. "CU": Clinical Understanding. "IP": Imaging Perception. "AR": Anatomical Recognition. "BS": Basic Science. "Diag.": Diagnosis. "Treat.": Treatment.

Model	Avg	CU	IP	AR	BS	Diag.	Treat.	Model	Avg	CU	IP	AR	BS	Diag.	Treat.
Proprietary General Models															
GPT-5	64.5(↓3.1)	67.2(↓4.3)	36.5(↓0.9)	67.4(↓4.4)	75.0(↓6.0)	62.1(↓2.9)	78.8(—-)	GPT-4o	47.5(↓2.8)	64.8(↓5.6)	32.0(—-)	61.9(↓4.3)	42.9(↑0.2)	37.4(↓3.9)	46.1(↓3.0)
Gemini-2.5-Pro	58.2(↓3.8)	66.9(↓3.4)	30.0(↓6.5)	72.0(↓5.7)	60.7(↓8.3)	60.1(↓1.3)	59.6(↑2.6)	GPT-4.1-mini	47.0(↓1.0)	65.9(↓1.9)	34.0(↓2.0)	63.5(↓1.8)	33.9(—-)	44.4(—-)	40.4(↓0.1)
Gemini-2.5-Flash	50.7(↓1.0)	63.6(↓4.8)	25.5(↓1.2)	71.2(↑0.1)	58.9(—-)	38.9(—-)	46.1(—-)	Claude-Sonnet-4.5	45.1(↓2.8)	62.8(↓3.8)	22.0(↓4.5)	44.3(↓6.3)	51.8(↑2.3)	35.9(↓4.6)	53.9(—-)
GPT-5.1	49.6(↓5.9)	65.9(↓3.3)	30.5(↓6.0)	65.8(↓4.2)	50.0(↓9.5)	47.0(↓8.5)	38.5(↓3.9)	GPT-4o-mini	31.6(↓2.7)	57.3(↓3.8)	24.5(↓6.7)	50.4(↓5.7)	12.5(↓0.1)	29.3(↓0.2)	15.4(—-)
GPT-4.1	48.9(↓1.0)	66.5(↓2.8)	32.5(↓1.9)	63.8(↓1.1)	50.0(—-)	38.4(—-)	42.3(—-)								
Open-Source General Models															
Qwen3-VL-Instruct(235B)	43.8(↓4.0)	73.0(↓3.7)	40.5(↓8.1)	66.8(↓4.9)	37.5(↓2.3)	29.8(↓4.9)	15.4(—-)	Qwen2-VL-Instruct(7B)	31.4(↓2.5)	61.7(↓2.8)	30.5(↓1.6)	39.7(↓3.0)	23.2(↑1.5)	19.7(↓3.3)	13.5(↓5.8)
InternVL3-Instruct(78B)	39.3(↓3.7)	75.9(↓3.1)	35.0(↑0.3)	66.3(↓3.1)	21.4(↓9.1)	29.3(↓7.0)	7.7(—-)	Idefics3(8B)	31.0(↓1.1)	60.0(↓3.0)	25.0(—-)	46.2(↓3.6)	17.9(—-)	25.2(—-)	11.5(↑0.2)
GLM-4.5v	38.5(↓3.2)	66.4(↓3.6)	25.5(—-)	57.7(↓6.3)	37.5(↓5.1)	32.3(↑0.1)	11.5(↓4.3)	Qwen2.5-VL-Instruct(32B)	29.9(↓1.4)	61.9(↓4.4)	27.5(↓0.1)	39.7(↓4.0)	17.9(↑0.1)	32.3(↓0.1)	0.0(↑0.1)
Qwen3-VL-Instruct(30B)	38.1(↓3.8)	73.0(↓2.6)	25.5(↓4.7)	66.1(↓8.1)	14.3(↓9.1)	30.3(↑1.8)	19.2(—-)	Qwen2-VL-Instruct(72B)	27.3(↓3.1)	62.8(↓3.3)	24.0(—-)	47.8(↓3.1)	5.4(—-)	20.2(↓0.9)	3.9(↓11.5)
InternVL2.5-MPO(38B)	36.5(↓1.2)	72.6(↓2.3)	30.0(↓0.1)	57.0(↓4.1)	25.0(—-)	22.7(↓0.5)	11.5(—-)	Janus-Pro(7B)	27.1(↓1.8)	50.7(↓1.5)	37.0(↓4.2)	32.7(↓2.7)	30.4(↓1.5)	11.6(↓1.0)	0.0(—-)
Mistral-Small-3.1-Instruct(24B)	33.9(↓2.2)	56.3(↓2.9)	25.0(↓0.3)	43.9(↓2.3)	25.0(—-)	33.8(—-)	19.2(↓7.7)	ShowO2(7B)	26.8(↓3.0)	58.6(↓4.0)	25.0(↓0.1)	39.2(↓8.2)	7.1(↑0.1)	21.2(↓0.1)	9.6(↓5.7)
InternVL3-Instruct(38B)	33.4(↓4.0)	71.7(↓3.0)	24.0(↓8.0)	58.3(↓5.8)	10.7(↓4.5)	31.8(↓2.6)	3.9(—-)	ShowO2(1.5B)	26.6(↓2.4)	46.3(↓2.6)	24.0(↓5.3)	33.2(↓1.2)	21.4(↓4.5)	15.7(↓0.5)	19.2(—-)
InternVL2.5(26B)	33.0(↓2.3)	71.8(↓2.8)	27.5(↓6.4)	55.7(↓4.5)	12.5(—-)	22.7(↓0.1)	7.7(↑0.1)	Qwen2.5-VL-Instruct(3B)	25.4(↓3.4)	48.7(↓5.0)	28.0(↓3.1)	33.6(↓4.5)	26.8(↓6.8)	15.2(↓0.8)	0.0(—-)
Qwen2.5-VL-Instruct(72B)	31.8(↓4.0)	63.4(↓4.3)	28.5(↓3.7)	44.5(↓5.7)	19.6(↓2.3)	28.8(↓5.9)	5.8(↓1.9)	BAGEL-MoT(7B)	24.7(↓2.2)	47.7(↓5.6)	32.5(—-)	35.4(↓7.9)	12.5(—-)	16.2(—-)	3.9(↑0.1)
Gemma-3(27B)	31.5(↓2.6)	58.7(↓4.9)	23.0(↓0.1)	55.2(↓6.1)	14.3(—-)	18.7(↓4.6)	19.2(↑0.1)	DeepSeek-VL2	23.2(↓1.4)	47.4(↓1.9)	29.5(↓2.8)	42.0(↓3.9)	7.1(—-)	1.5(↑0.3)	11.5(—-)
Gemma-3(4B)	31.5(↓4.2)	54.4(↓2.6)	32.5(↓0.1)	43.2(↓7.2)	17.9(↓15.0)	14.1(—-)	26.9(↑0.1)								
Medical-specialized Models															
UniMedVL	42.5(↓3.7)	69.2(↓3.6)	40.0(↓3.7)	63.1(↓5.6)	25.0(—-)	32.8(↓7.6)	25.0(↓1.9)	HealthGPT-L(14B)	35.2(↓1.3)	66.6(↓1.5)	33.0(↓4.6)	45.3(↓4.3)	17.9(—-)	40.9(↑2.8)	7.7(—-)
Hulu-Med(7B)	38.8(↓2.2)	71.0(↓4.2)	31.0(↑1.0)	57.5(↓5.0)	28.6(↓3.4)	27.3(—-)	17.3(↓1.9)	MedGemma(4B)	35.0(↓1.9)	62.1(↓2.9)	30.0(↓1.9)	48.0(↓2.5)	17.9(↑2.3)	26.8(↓0.6)	25.0(↓5.8)
Hulu-Med(14B)	38.4(↓1.5)	72.8(↓2.1)	34.0(↓2.3)	64.0(↓4.6)	26.8(↓0.1)	25.2(↓0.1)	7.7(↑0.1)	HealthGPT-M(3B)	32.6(↓1.6)	63.7(↓2.6)	28.5(↓0.8)	42.0(↓1.9)	14.3(↓3.7)	27.8(↓0.7)	19.2(—-)
Lingshu(32B)	36.1(↓3.9)	70.4(↓3.8)	27.5(↑0.5)	56.2(↓8.7)	26.8(↓8.3)	27.8(↓2.9)	7.7(—-)	MedGemma(27B)	31.4(↓2.2)	59.3(↓4.1)	26.5(↓0.1)	38.0(↓3.5)	10.7(↑0.1)	42.4(↓5.6)	11.5(—-)
Lingshu(7B)	35.8(↓4.6)	71.7(↓4.4)	23.5(↓8.6)	56.3(↓8.1)	23.2(—-)	20.7(↓6.3)	19.2(—-)	HealthGPT-XL(32B)	30.5(↓3.1)	61.4(↓3.9)	33.0(↓0.3)	48.5(↓2.6)	17.9(↓9.1)	14.7(↓2.6)	7.7(—-)
Average (All Models)		**63.3**(↓3.4)	**29.4**(↓2.4)	**52.1**(↓4.5)	**26.0**(↓2.8)	**28.6**(↓1.9)	**19.2**(↓1.6)								

5 (Singh et al., 2025), GPT-5.1 (OpenAI, 2025b), GPT-4o-mini, GPT-4o (Hurst et al., 2024), GPT-4.1-mini, GPT-4.1 (OpenAI, 2025a), Gemini-2.5-Pro (Comanici et al., 2025), Gemini-2.5-Flash (Comanici et al., 2025) and Claude-Sonnet-4.5 (Anthropic, 2025);

(2) **21** ***Open-source general models***, including InternVL3-Instruct (78B, 38B) (Zhu et al., 2025), InternVL2.5-MPO (38B) (Chen et al., 2025b), InternVL2.5 (26B) (Chen et al., 2025b), Qwen3-VL-Instruct (235B, 30B) (Bai et al., 2025a), Qwen2.5-VL-Instruct (72B, 32B, 3B) (Bai et al., 2025b), Qwen2-VL-Instruct (72B, 7B) (Wang et al., 2024), Gemma-3 (27B, 4B) (Team et al., 2025a), Idefics3 (8B) (Laurençon et al., 2024), Mistral-Small-3.1-Instruct (24B) (AI, 2025), ShowO2 (7B, 1.5B) (Xie et al., 2025), BAGEL-MoT (7B) (Deng et al., 2025), DeepSeek-VL2 (Wu et al., 2024), GLM-4.5v (Team et al., 2026), and Janus-Pro (7B) (Chen et al., 2025a).

(3) **10** ***Medical-specialized models***, including UniMedVL (Ning et al., 2025), Hulu-Med (14B, 7B) (Jiang et al., 2025), Lingshu (32B, 7B) (Team et al., 2025b), HealthGPT-XL/L/M (32B, 14B, 3B) (Lin et al., 2025), and MedGemma (27B, 4B) (Sellergren et al., 2025).

For each VQA pair, each MLLM is tested across three severity degrees (L0, L1, L2). Degradations are implemented using TorchIO[1], Torch Radon[2], SciPy ndimage[3], and Froodo[4]. To ensure statistically significant assessments of model con-

[1] https://docs.torchio.org/
[2] https://torch-radon.readthedocs.io/
[3] https://docs.scipy.org/doc/scipy/reference/ndimage.html
[4] https://github.com/iserh/froodo

Table 3. **Degradation-category performance (L1/L2) with degradation magnitude.** Average accuracy across five degradation categories, computed over L1 and L2 degradations. Numbers in parentheses show performance drop from L0 to L1&L2. "Intens.": intensity jitter. "R&B": resolution and blur issues. "Artif.": artifacts. "Motio.": motion interference.

Model	Avg	Intens.	Noise	R&B	Artif.	Motio.	Model	Avg	Intens.	Noise	R&B	Artif.	Motio.
Proprietary General Models													
Gemini-2.5-Pro	67.6(↓4.1)	71.1(↓0.9)	69.2(↓2.3)	67.2(↓2.0)	65.1(↓5.3)	68.6(↓8.0)	GPT-4.1-mini	63.8(↓1.8)	67.1(↓1.6)	63.3(↑0.8)	64.4(↓1.4)	60.5(↓3.1)	66.8(↓2.5)
GPT-5	66.7(↓4.2)	72.2(↓4.0)	66.2(↓3.2)	67.5(↓2.5)	63.1(↓6.4)	68.4(↓4.8)	GPT-4o	62.5(↓5.0)	68.8(↓1.3)	61.8(↓5.5)	63.2(↓4.6)	58.3(↓6.7)	65.0(↓4.2)
GPT-5.1	64.8(↓4.6)	70.9(↓2.5)	64.2(↓4.8)	65.7(↓4.2)	60.5(↓6.2)	67.2(↓3.4)	GPT-4o-mini	53.5(↓4.4)	62.3(↓2.7)	53.9(↓4.3)	55.1(↓3.2)	47.8(↓6.0)	54.2(↓6.1)
Gemini-2.5-Flash	64.6(↓3.0)	67.5(↓0.6)	65.1(↓1.8)	64.8(↓1.0)	60.7(↓5.0)	68.7(↓4.0)	Claude-Sonnet-4.5	55.6(↓4.6)	59.1(↓2.7)	53.3(↓5.6)	56.6(↓4.5)	55.2(↓7.0)	54.3(↓5.5)
GPT-4.1	64.2(↓2.2)	69.4(↓3.7)	62.7(↓0.8)	65.1(↓2.4)	60.6(↓3.2)	67.4(↓0.6)							
Open-Source General Models													
InternVL3-Instruct(78B)	70.5(↓3.1)	75.1(↓3.6)	69.3(↓2.8)	72.7(↓2.0)	64.3(↓5.2)	75.6(↓2.5)	Qwen2.5-VL-Instruct(32B)	53.2(↓4.0)	50.6(↓4.3)	51.8(↓4.2)	54.2(↓2.5)	52.3(↓7.2)	56.0(↓4.7)
Qwen3-VL-Instruct(235B)	69.1(↓4.2)	75.3(↓2.9)	69.8(↓2.7)	70.7(↓2.2)	63.3(↓6.2)	71.1(↓6.5)	Qwen2-VL-Instruct(7B)	53.0(↓2.9)	53.7(↓1.2)	52.6(↓2.7)	53.1(↓2.2)	53.0(↓4.5)	52.5(↓4.3)
Qwen3-VL-Instruct(30B)	68.5(↓4.3)	76.2(↓1.0)	68.0(↓4.5)	71.3(↓2.2)	61.4(↓7.1)	70.3(↓6.4)	Mistral-Small-3.1-Instruct(24B)	51.0(↓2.6)	52.4(↓0.2)	49.0(↓2.8)	51.8(↓0.5)	49.5(↓5.5)	53.7(↓3.8)
InternVL2.5-MPO(38B)	65.3(↓2.8)	70.1(↓1.4)	64.2(↓1.7)	66.5(↓2.4)	61.9(↓4.4)	67.4(↓3.1)	ShowO2(7B)	50.7(↓5.1)	62.3(↓1.6)	47.0(↓6.3)	52.8(↓5.9)	45.2(↓4.5)	52.6(↓6.1)
InternVL3-Instruct(38B)	65.1(↓4.0)	70.5(↓1.6)	64.0(↓4.3)	67.4(↓2.7)	59.9(↓6.2)	67.2(↓4.1)	Gemma-3(4B)	49.4(↓4.0)	54.7(↓1.9)	49.9(↓2.4)	50.1(↓5.3)	46.0(↓5.0)	49.9(↓5.6)
InternVL2.5(26B)	64.3(↓3.4)	73.6(↓3.2)	62.6(↓1.9)	66.3(↓3.0)	59.9(↓4.1)	63.7(↓5.3)	Janus-Pro(7B)	43.7(↓1.9)	53.7(↓0.5)	41.2(↓3.7)	46.9(↓1.3)	38.4(↓2.9)	42.7(↓0.5)
GLM-4.5v	61.8(↓4.3)	65.5(↓1.7)	63.9(↓0.6)	62.6(↓3.8)	57.2(↓7.4)	63.0(↓9.4)	Qwen2.5-VL-Instruct(3B)	42.6(↓4.7)	48.0(↓3.5)	39.6(↓6.0)	43.0(↓4.0)	39.3(↓7.0)	47.7(↓0.7)
Qwen2-VL-Instruct(72B)	55.9(↓3.1)	58.5(↓3.5)	54.8(↓2.4)	57.5(↓2.5)	53.2(↓5.4)	57.1(↓3.5)	DeepSeek-VL2	44.1(↓2.5)	33.0(↓1.1)	46.9(↓2.5)	43.0(↓1.9)	45.0(↓3.6)	48.6(↓3.4)
Gemma-3(27B)	55.7(↓5.1)	58.2(↓1.7)	58.7(↓1.8)	55.2(↓5.7)	51.4(↓7.0)	59.1(↓5.6)	BAGEL-MoT(7B)	42.5(↓6.0)	53.9(↓2.9)	39.9(↓8.2)	44.0(↓5.8)	37.0(↓5.5)	45.1(↓5.6)
Qwen2.5-VL-Instruct(72B)	55.6(↓4.7)	57.6(↓3.4)	53.0(↓5.1)	57.2(↓3.7)	53.6(↓7.5)	57.5(↓4.1)	ShowO2(1.5B)	40.9(↓2.1)	47.8(↓1.3)	38.9(↓1.7)	42.4(↓2.0)	37.8(↓2.0)	41.0(↓1.8)
Idefics3(8B)	53.8(↓3.0)	65.7(↓0.9)	50.5(↓4.7)	55.7(↓3.1)	48.0(↓2.3)	56.6(↓3.6)							
Medical-specialized Models													
Hulu-Med(14B)	67.8(↓2.8)	70.2(↓2.3)	66.3(↓2.2)	69.7(↓1.8)	64.0(↓4.9)	70.6(↓4.0)	HealthGPT-L(14B)	58.2(↓2.4)	63.5(↓2.0)	57.0(↓1.6)	59.5(↓1.8)	55.3(↓4.1)	58.8(↓3.4)
UniMedVL	65.6(↓4.3)	76.1(↓0.9)	65.3(↓2.4)	66.1(↓4.2)	60.1(↓6.5)	67.8(↓5.0)	MedGemma(4B)	55.9(↓2.7)	66.0(↓1.8)	54.8(↓1.9)	59.2(↓2.5)	49.3(↓5.8)	55.5(↓4.5)
Hulu-Med(7B)	64.6(↓4.2)	69.9(↓1.1)	63.5(↓3.1)	65.8(↓3.8)	61.1(↓6.1)	66.5(↓5.3)	HealthGPT-XL(32B)	55.4(↓3.4)	62.2(↓2.1)	55.0(↓2.6)	56.1(↓2.0)	51.7(↓6.4)	56.7(↓4.9)
Lingshu(7B)	64.4(↓5.7)	73.7(↓3.1)	62.2(↓3.7)	65.9(↓4.3)	60.0(↓5.2)	65.6(↓10.1)	HealthGPT-M(3B)	55.0(↓2.3)	58.9(↓1.4)	54.7(↓0.8)	55.8(↓1.6)	52.3(↓4.3)	56.0(↓2.5)
Lingshu(32B)	63.7(↓5.3)	74.0(↓1.9)	62.1(↓3.8)	65.1(↓4.3)	57.7(↓6.4)	66.7(↓8.2)	MedGemma(27B)	51.1(↓3.8)	62.2(↓0.4)	47.4(↓5.1)	52.3(↓2.3)	48.8(↓5.2)	50.2(↓5.4)
Average (All Models)		**63.4**(↓2.1)	**57.1**(↓3.2)	**59.2**(↓2.9)	**54.1**(↓5.4)	**59.8**(↓4.6)							

fidence and uncertainty, we perform $T = 3$ independent inference trials for every test sample and $T = 10$ for calibration shift experiment. All models are evaluated with temperature 1.0 to obtain unbiased confidence estimates.

3.2. Findings on Severity Degrees

Conclusion 1. Most evaluated MLLMs exhibit severe robustness deficiency with nonlinear degradation patterns. As shown in Figure 4 and Table 6 (Appendix), the majority of the 40 MLLMs suffer significant performance drops as image quality decreases. Notably, even the best-performing model InternVL3-Instruct (78B) across all levels experiences substantial accuracy drops at L2, showing that robustness deficiency remains a widespread challenge across both proprietary and open-source mdoels.

Beyond a simple monotonic decline, the performance degradation exhibits a concerning nonlinear acceleration pattern. While many models show some tolerance from L0 to L1 severity degrees, the performance deterioration rate increases sharply from L1 to L2. This suggests a "cliff effect" where model perception remains relatively stable until a certain threshold of noise is reached, after which the vision-language integration undergoes a catastrophic collapse.

3.3. Findings on Capability Dimensions

We analyze robustness across the six mid-level capability dimensions. Table 2 provides comprehensive performance.

Conclusion 2. Models exhibit dramatic performance disparities across capability dimensions, with reasoning tasks being far more fragile than perception. Clinical Understanding (CU) remains the strongest capability for most models, while Imaging Perception (IP), Basic Science reasoning (BS), Diagnosis (Diag), and Treatment planning (Treat) demonstrate critically weak performance. Notably, even top-performing models like InternVL3-Instruct (78B) and GPT-5, which excel at factual knowledge recall, suffer substantial drops in clinical reasoning dimensions.

Further, medical-specialized models show comparable or worse performance than general models across most dimensions, with particularly catastrophic failures in treatment planning where many models collapse to near-zero accuracy. Furthermore, degradation-induced performance drops are substantially larger in clinical reasoning tasks compared to factual knowledge tasks, indicating that these higher-order cognitive capabilities are more vulnerable to image quality degradation. This pattern reveals that current MLLMs possess superficial medical knowledge but fundamentally lack the integrative reasoning and perceptual capabilities essential for clinical decision-making.

3.4. Findings on Degradation-Type Sensitivity

We analyze model sensitivity to different degradation types. Comprehensive degradation heatmaps showing performance drops from baseline (L0) to degraded levels (L1 and L2) across different models and degradation types are presented in Table 3.

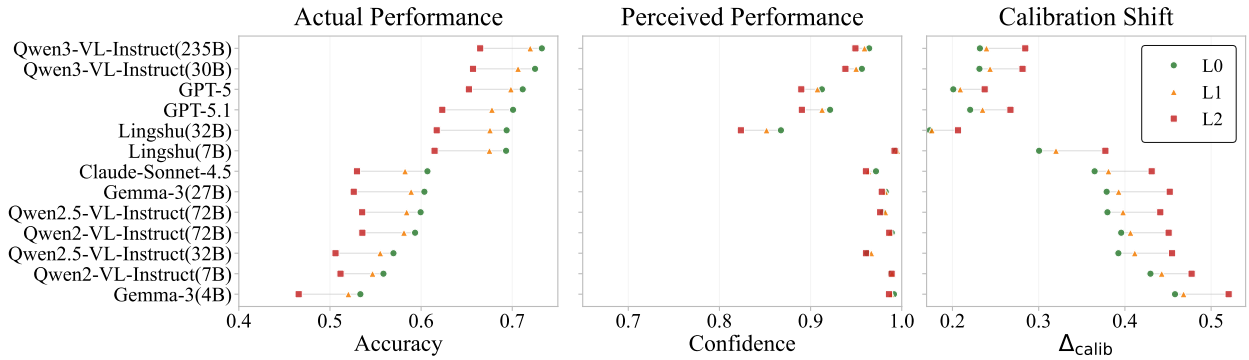

Figure 5. **Comprehensive analysis of model calibration shift across severity degrees.** These panels quantify the widening gap between actual accuracy and model certainty under increasing image degradation. As degradation scales from L0 to L2, the stability of perceived performance against collapsing accuracy reveals a fundamental failure in metacognitive awareness across MLLMs.

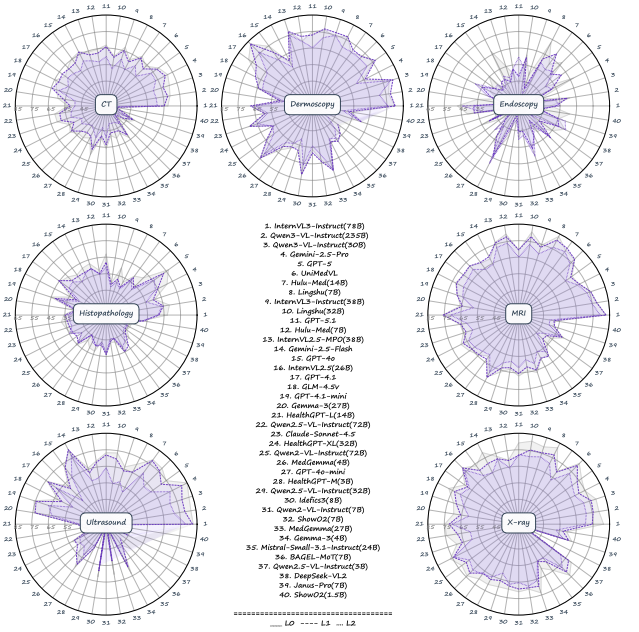

Figure 6. **Performance comparison across seven medical imaging modalities under three severity degrees.** Each radar chart shows model performance (accuracy) for a specific modality. Zoom in for better viewing.

Conclusion 3. MLLMs are strikingly vulnerable to physics-based artifacts and motion interference compared to intensity-based degradations. The five degradation categories produce markedly different impacts on model robustness. Intensity degradations demonstrate the highest resilience with the smallest performance drops, while noise and resolution & blur show moderate impact. In sharp contrast, artifacts degradations, which encompass medical physics-specific corruptions such as MRI motion artifacts, CT metal artifacts, and pathology staining variations, cause larger drops than intensity degradations. Motion-induced degradations also show substantial impact, ranking second in severity after artifacts. This vulnerability hierarchy reveals that current MLLMs lack sufficient understanding of medical imaging-specific degradation mechanisms fundamentally absent from natural image pretraining distributions, and that degradations disrupting semantic medical content (artifacts) and spatial structure (motion interference) are substantially more destructive than photometric variations (intensity, resolution & blur). Notably, top-performing models exhibit significantly larger performance losses on these physics-based corruptions than on basic intensity adjustments.

3.5. Findings on Modality-Specific Performance

Figure 6 presents a comprehensive view of model performance across seven major medical imaging modalities.

Conclusion 4. Models exhibit striking modality-specific robustness patterns, with Endoscopy and Histopathology showing particularly severe vulnerability. As shown in Figure 6, different medical imaging modalities present vastly different robustness challenges. Endoscopy and Histopathology consistently show lower performance across all severity levels compared to CT, MRI, Ultrasound, and X-ray. This disparity reflects fundamental differences in imaging characteristics: Histopathology requires interpretation of complex tissue microstructure and staining patterns highly sensitive to color variations, while Endoscopy presents challenges from variable illumination, reflections, and limited field-of-view. In contrast, cross-sectional imaging modalities (CT, MRI) and Ultrasound maintain relatively higher robustness, likely benefiting from standardized acquisition protocols and models' better exposure to quality shift during training.

3.6. Findings on Overconfidence Under Degradations

We analyze the critical disconnect between actual performance and perceived confidence under degradation. Follow-

ing the Calibration Shift methodology 2.4, we measure the gap between perceived confidence and actual performance.

Conclusion 4. All models exhibit severe overconfidence under degradation, demonstrating the AI Dunning-Kruger Effect. As shown in Figure 5, models maintain inappropriately high perceived confidence despite substantial accuracy degradation. Across all models and quality levels, calibration shift remains consistently positive and increases systematically from L0 to L2. This systematic widening of the capability-confidence gap precisely characterizes the Dunning-Kruger Effect, where decreased competence paradoxically correlates with inflated self-assessment.

Our analysis reveals this phenomenon manifests in two complementary dimensions. **Intra-Model DKE** characterizes metacognitive failure within individual models: as degradation severity increases from L0 to L2, actual performance drops substantially while calibration shift consistently increases, indicating models grow more overconfident precisely when their capabilities deteriorate. This pattern holds universally across all evaluated architectures. **Inter-Model DKE** describes the inverse relationship across different models: at severe degradation levels, lower-performing models exhibit disproportionately higher calibration shift compared to higher-capability models.

4. Related Work

4.1. Medical Multimodal Large Language Models

The application of Multimodal Large Language Models in medical domains has witnessed rapid advancement. To better address medical-specific challenges, a series of *medical-specialized models* have emerged, including GMAI-VL-R1 (Su et al., 2025), UniMedVL (Ning et al., 2025), Hulu-Med (Jiang et al., 2025), Lingshu (Team et al., 2025b), HealthGPT (Lin et al., 2025), and MedGemma (Sellergren et al., 2025). These models are typically pre-trained or fine-tuned on large-scale medical image-text datasets to enhance domain-specific understanding.

To systematically evaluate these medical MLLMs, numerous benchmarks have been developed targeting various medical scenarios. GMAI-MMBench (Ye et al., 2024) integrates multiple medical datasets covering clinical tasks, department categorization, and perception granularity. OmniMed-VQA (Hu et al., 2024) aggregates various medical VQA datasets to provide comprehensive evaluation. Modality-specific benchmarks include PathVQA (He et al., 2020) for pathology images and VQA-RAD (Lau et al., 2018) for radiology. SLAKE (Liu et al., 2021) provides bilingual medical VQA with segmentation masks. General multimodal benchmarks such as MMMU (Yue et al., 2024) and MMMU-Pro (Yue et al., 2025) encompass multidisciplinary tasks including medical content. Recent benchmarks MedXpertQA (Zuo et al., 2025) for expert-level exam-style question answering. Despite their comprehensive coverage, these benchmarks primarily evaluate performance on standard clean data without systematically considering robustness to image degradations that commonly occur in real-world clinical environments.

4.2. Evaluating MLLMs Under Image Degradations

In computer vision, ImageNet-C (Hendrycks & Dietterich, 2019) pioneered systematic evaluation by introducing 15 common corruption types with 5 severity levels for image classification. This framework has been extended to evaluate MLLMs under various image quality conditions (Li et al., 2025a; Qiu et al., 2025), revealing that even state-of-the-art models exhibit significant performance degradation when confronted with corrupted natural images.

Recent benchmarks have begun examining MLLM resilience to input variations, including Res-Bench (Sui et al., 2025) for dynamic resolution robustness and discriminative corruption robustness assessment (Li et al., 2025a). Some works expand the evaluation to medical MLLM behavior under image perturbations. MedQBench (Liu et al., 2025) evaluates MLLMs' capability in assessing medical image quality. Xu *et al.* (XU et al., 2025) and Cheng *et al.* (Cheng et al., 2025) investigated medical MLLM responses to text and image perturbations and proposed enhancement methods. However, existing evaluations remain limited in scope, often focusing on single modalities or specific perturbation types without comprehensive assessment across diverse medical tasks and degradation categories.

5. Conclusion

This paper presents MedQ-Deg, a systematic benchmark for evaluating medical MLLM performance under image quality variations. Through comprehensive evaluation across capability dimensions, degradation categories, and severity levels, we reveal severe vulnerabilities in current models when confronted with degraded medical images. Our key findings show universal performance degradation as image quality decreases. Most critically, we demonstrate the AI Dunning-Kruger Effect: models exhibit persistent overconfidence despite accuracy collapse, posing grave safety risks for clinical deployment. These findings underscore that reliable performance under image quality variations cannot be an afterthought but must be fundamental to medical MLLM development. We believe MedQ-Deg provides essential infrastructure for advancing this critical research direction.

Impact Statement

This work introduces the MedQ-Deg benchmark, constructed from publicly available medical imaging datasets with no protected health information included. The benchmark is intended exclusively for research evaluation rather than clinical decision-making. By establishing systematic robustness evaluation standards, this work advances medical imaging models toward more reliable and trustworthy AI systems under appropriate supervision.

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A. Complete Capability Dimension Hierarchy

Name	Parent Category	Count	Ratio in Parent	Ratio in Total
Skeletal System	Anatomical Recognition	2,562	41.22%	12.45%
Cardiovascular	Anatomical Recognition	1,392	22.39%	6.76%
Abdominal Structures	Anatomical Recognition	880	14.16%	4.27%
Respiratory System	Anatomical Recognition	726	11.68%	3.53%
Gastrointestinal Tract	Anatomical Recognition	475	7.64%	2.31%
Neurovascular	Anatomical Recognition	463	7.45%	2.25%
General Structures	Anatomical Recognition	328	5.28%	1.59%
Muscle Recognition	Anatomical Recognition	231	3.72%	1.12%
Cell and Microbes	Anatomical Recognition	158	2.54%	0.77%
General Science Reasoning	Basic Science Reasoning	74	71.15%	0.36%
Pathophysiological Reasoning	Basic Science Reasoning	26	25.00%	0.13%
Clinical Correlation	Basic Science Reasoning	4	3.85%	0.02%
Disease Diagnosis	Clinical Understanding	12,908	97.74%	62.70%
Severity Grading	Clinical Understanding	138	1.04%	0.67%
Basic Science Understanding	Clinical Understanding	78	0.59%	0.38%
Lesion Grading	Clinical Understanding	68	0.51%	0.33%
Attribute Recognition	Clinical Understanding	10	0.08%	0.05%
Treatment-Understanding	Clinical Understanding	4	0.03%	0.02%
General Diagnosis Reasoning	Diagnosis Reasoning	336	87.50%	1.63%
Laboratory/Imaging Interpretation	Diagnosis Reasoning	38	9.90%	0.18%
Clinical Presentation Analysis	Diagnosis Reasoning	6	1.56%	0.03%
Disease Mechanism Recognition	Diagnosis Reasoning	4	1.04%	0.02%
Counting	Imaging Perception	390	73.58%	1.89%
Surgical Workflow Recognition	Imaging Perception	96	18.11%	0.47%
Image Quality Grading	Imaging Perception	28	5.28%	0.14%
Surgical Instrument Recognition	Imaging Perception	16	3.02%	0.08%
Treatment Reasoning	Treatment Reasoning	127	85.81%	0.62%
Pharmacological Treatment	Treatment Reasoning	12	8.11%	0.06%
Initial/Emergency Management	Treatment Reasoning	7	4.73%	0.03%
Surgical Treatment Selection	Treatment Reasoning	2	1.35%	0.01%

Table 4. **Fine-grained capability distribution.**

B. Complete Degradation Taxonomy

Name	Parent Category	Count	Ratio in Parent	Ratio in Total	Modality
Artifacts	-	**5,322**	-	**25.85%**	**All**
limited_angle	Artifacts	1,440	27.06%	6.99%	CT
sparse_view	Artifacts	1,440	27.06%	6.99%	CT
bias_field_artifact	Artifacts	936	17.59%	4.55%	MRI
undersampling_artifact	Artifacts	672	12.63%	3.26%	MRI
ghosting_artifact	Artifacts	352	6.61%	1.71%	MRI
blood_cell_artifact	Artifacts	304	5.71%	1.48%	Pathology
dark_spots_artifact	Artifacts	178	3.34%	0.86%	Pathology
Motion	-	**2,992**	-	**14.53%**	**All**
object_rotation	Motion	2,718	90.84%	13.20%	All
object_movement	Motion	274	9.16%	1.33%	All
Intensity	-	**1,846**	-	**8.97%**	**All**
adjust_brightness	Intensity	916	49.62%	4.45%	All
exposure	Intensity	518	28.06%	2.52%	All
reduce_contrast	Intensity	412	22.32%	2.00%	All
Noise	-	**3,796**	-	**18.44%**	**All**
gaussian_noise	Noise	2,680	70.60%	13.02%	All
low_dose	Noise	1,116	29.40%	5.42%	CT
Resolution and Blur	-	**6,632**	-	**32.21%**	**All**
low_resolution	Resolution and Blur	3,046	45.93%	14.80%	All
motion_blur	Resolution and Blur	2,820	42.52%	13.70%	All
gaussian_blur	Resolution and Blur	470	7.09%	2.28%	All
bubble	Resolution and Blur	296	4.46%	1.44%	Pathology

Table 5. **Degradation type statistics.**

C. Detail Results

Model	L0↑	L1↑	L2↑	L1-L0↓	L2-L0↓
Gemini-2.5-Pro	71.39	69.93	65.20	-1.46	-6.19
GPT-5	70.27	69.09	64.30	-1.18	-5.97
GPT-5.1	68.56	67.34	62.16	-1.22	-6.40
Gemini-2.5-Flash	67.39	67.18	62.00	-0.21	-5.39
GPT-4o	66.98	65.22	59.73	-1.76	-7.25
GPT-4.1	66.00	66.50	62.00	0.50	-4.00
GPT-4.1-mini	65.21	65.87	61.67	0.66	-3.54
Claude-Sonnet-4.5	59.64	58.28	52.86	-1.36	-6.78
GPT-4o-mini	57.32	55.49	51.53	-1.83	-5.79
InternVL3-Instruct(78B)	74.04	72.46	68.63	-1.58	-5.41
Qwen3-VL-Instruct(235B)	73.15	71.87	66.40	-1.28	-6.75
Qwen3-VL-Instruct(30B)	72.74	70.79	66.14	-1.95	-6.60
InternVL3-Instruct(38B)	68.83	67.38	62.87	-1.45	-5.96
InternVL2.5-MPO(38B)	68.00	66.87	63.81	-1.13	-4.19
InternVL2.5(26B)	66.88	66.48	62.12	-0.40	-4.76
GLM-4.5v	65.77	64.57	58.94	-1.20	-6.83
Gemma-3(27B)	61.03	58.91	52.45	-2.12	-8.58
Qwen2.5-VL-Instruct(72B)	59.68	58.28	52.92	-1.40	-6.76
Qwen2-VL-Instruct(72B)	58.52	58.26	53.61	-0.26	-4.91
Qwen2.5-VL-Instruct(32B)	57.22	55.67	50.69	-1.55	-6.53
Idefics3(8B)	56.71	55.85	51.82	-0.86	-4.89
Qwen2-VL-Instruct(7B)	55.69	54.77	51.14	-0.92	-4.55
ShowO2(7B)	55.60	53.59	47.72	-2.01	-7.88
Gemma-3(4B)	53.60	52.06	46.70	-1.54	-6.90
Mistral-Small-3.1-Instruct(24B)	53.55	53.43	48.61	-0.12	-4.94
BAGEL-MoT(7B)	48.30	45.17	39.92	-3.13	-8.38
Qwen2.5-VL-Instruct(3B)	46.86	44.72	40.43	-2.14	-6.43
DeepSeek-VL2	46.77	45.37	42.80	-1.40	-3.97
Janus-Pro(7B)	45.29	45.23	42.17	-0.06	-3.12
ShowO2(1.5B)	43.15	42.47	39.35	-0.68	-3.80
UniMedVL	70.18	68.48	62.69	-1.70	-7.49
Hulu-Med(14B)	70.04	70.20	65.40	0.16	-4.64
Lingshu(7B)	69.67	67.48	61.34	-2.19	-8.33
Lingshu(32B)	68.65	67.07	60.40	-1.58	-8.25
Hulu-Med(7B)	68.09	67.07	62.20	-1.02	-5.89
HealthGPT-L(14B)	60.29	59.71	56.75	-0.58	-3.54
HealthGPT-XL(32B)	58.99	57.16	53.69	-1.83	-5.30
MedGemma(4B)	58.48	58.12	53.77	-0.36	-4.71
HealthGPT-M(3B)	57.32	56.61	53.45	-0.71	-3.87
MedGemma(27B)	54.90	53.06	49.22	-1.84	-5.68

Table 6. **Overall accuracy across severity levels.** L0/L1/L2 accuracies and drops (L1-L0, L2-L0); rows are grouped by model family.